Reg. No. | | | | | | | | | | | [14EE663]

SIXTH SEMESTER B. Tech EXAMINATION, JUNE–2019

ELECTRONIC INSTRUMENTATION

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions. choosing by Atleast two questions from each part.
2. Missing data may be suitably assumed.

PART - A

1.	a.	What are the different types of errors in measurements? Explain them briefly.	07
	b.	A voltmeter having a sensitivity of 1 K ohm /volt is connected across an unknown resistance in series with a milli-ammeter reading 80V on 150 volts scale. when the milli ammeter reads 10mA.calculate the; i. apparent resistance of the unknown resistance ii. actual resistance of unknown resistance iii. Error due to the loading effect of the voltmeter.	07
2	a.	Explain the working principle of the following: i. Series type ohmmeter ii. Integrating type DVM	08
	b.	Draw & explain the block diagram of True RMS voltmeter.	06
3	a.	With a neat diagram, explain the following: i. Digital voltmeter ii. MP based instrument	08
	b.	How digital tachometer is different from conventional tachometer, mention its advantages.	06
4	a.	Explain Strip chart recorder with diagram.	07
	b.	Explain with suitable diagram of Null type recorder.	07
5	a.	With diagram explain the working of function generator, mention its advantages.	08
	b.	Write a note on Random noise generator.	06
6	a.	Explain the measurement Resistance by Kelvin's bridge. What range of Resistance can be measured by their bridge? Obtain the balance equations	07
	b.	What is the wagner's earth device? Why it is used? Explain its connection & operation in bridge balance.	07

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7	a	Define electrical transducers and mention the types. Explain their basic principle with relevant sketches.	07
	b	Draw & explain LVDT.	07
8	a	Write a note on classification of displays.	04
	b	Discuss on following display devices; i. LED ii. LCD	06
	c.	What are digital indicators? Discuss briefly.	04